

E_Lecture3_W1

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Q1 [30 pts, 10 pts for each]

如果 X 商品 需求函數 為下式

$$Q_X^D = 250 - 0.01P_X + 0.07 P_Y + 0.02I - 0.38R$$

其中 P_X 、 P_Y 為 X、Y 商品的價格， I 為所得；

如果原本 $P_X = 100$ ， $P_Y = 100$ ， $I = 1000$ ， $R = 0$ ，則

- (1) 當 P_X 上漲至 200 時，其他條件不變，需求價格彈性是多少？
- (2) 當 P_Y 上漲至 200 時，其他條件不變，需求交叉彈性是多少？
- (3) 當 I 上漲至 1500 時，其他條件不變，需求所得彈性是多少？

If demand curve for product X is

$$Q_X^D = 250 - 0.01P_X + 0.07 P_Y + 0.02I - 0.38R,$$

where P_X and P_Y are prices of product X and Y， I is income.

If the initial value of them is $P_X = 100$ ， $P_Y = 100$ ， $I = 1000$ ， $R = 0$.

- (1) As P_X increases from 100 to 200 and other things being equal, please calculate the price elasticity of demand for the product.
- (2) As P_Y increases from 100 to 200 and other things being equal, please calculate the cross elasticity of demand for the product.
- (3) As I increases from 1000 to 1500 and other things being equal, please calculate the income elasticity of demand for the product.

(1)

$$Q_1 = 250 - 1 + 7 + 20 = 276$$

$$Q_2 = 250 - 2 + 7 + 20 = 275$$

$$E_{X^D}^{P_X} = \left| \frac{(275-276)/275.5}{(200-100)/150} \right| = 0.0054$$

(2)

$$Q_1 = 250 - 1 + 7 + 20 = 276$$

$$Q_2 = 250 - 1 + 14 + 20 = 283$$

$$E_{X^D}^{P_Y} = \frac{(283-276)/279.5}{(200-100)/150} = 0.038$$

	(3) $Q_1 = 250 - 1 + 7 + 20 = 276$ $Q_2 = 250 - 1 + 7 + 30 = 286$ $E_x^{D.I} = ((286-276)/281)/((1500-1000)/1250) = 0.089$
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Q2	[40 pts, 10 pts for each] (1) A life-saving medicine without any close substitutes will tend to have a) a small elasticity of demand. b) a large elasticity of demand. c) a small elasticity of supply. d) a large elasticity of supply. Answer: <u> c </u> (2) The price of a good rises from \$8 to \$12, and the quantity demanded falls from 110 to 90 units. Calculated with the midpoint method, the price elasticity of demand is a) 1/5. b) 1/2. c) 2. d) 5. Answer: <u> b </u> (3) A linear, downward-sloping demand curve is a) inelastic. b) unit elastic.
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	c) elastic. d) inelastic at some points, and elastic at others. Answer: <u> b </u>
	(4) The ability of firms to enter and exit a market over times means that, in the long run, a) the demand curve is more elastic. b) the demand curve is less elastic. c) the supply curve is more elastic. d) the supply curve is less elastic Answer: <u> c </u>

Q3

[30 pts, 15 pts for each]

Suppose that business travelers and vacationers have the following demand for airline tickets from New York to Boston:

Price	Quantity Demanded (business travelers)	Quantity Demanded (vacationers)
\$150	2,100 tickets	1,000 tickets
200	2,000	800
250	1,900	600
300	1,800	400

(a)

As the price of tickets rises from \$200 to \$250, what is the price elasticity of demand for (i) business travelers and (ii) vacationers?

(b)

Why might vacationers have a different elasticity from business travelers?

(a)

$$E_{bt}^{D.P_{bt}} = \left| \frac{(1900-2000)/1950}{(250-200)/225} \right| = 0.23$$

$$E_v^{D.P_v} = \left| \frac{(600-800)/700}{(250-200)/225} \right| = 1.29$$

(b)

Under the same changes of price, the higher the percentage of Demand,

	the higher the Elasticity. Usually vacationers have more substitutes, and the more substitutes, the more elasticity
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